

MEETING INDUSTRY – ACADEMIA RESEARCH COLLABORATION CHALLENGES WITH AGILE METHODOLOGIES

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Abstract— Continuous and long-term collaboration between industry and academia is crucial to develop front-line research in context-dependent areas like software development where both practitioners and researchers are searched for data collection, analysis and results. Despite many mutual benefits, this collaboration is often challenging, not only due to different goals, but also because of different pace in providing the results. The software development industry has during the last decade aligned around and organized their development adopting agile methodologies. For the researchers, the agile methodologies are a topic for a research, rather than a means of performing the research itself. We can state a question, whether the agile methodologies can be a good common ground for enabling successful research collaboration between industry and academia? This paper reports on a longitudinal industry – academia research collaboration case, which has stepwise adapted SCRUM over a six-year period. The implementation of SCRUM and the collaboration successes and challenges are presented, and findings are discussed.

Keywords: industry-academia research; agile methodologies; collaboration; case study

I. INTRODUCTION

Speed and responsiveness are today essential for being successful in software development as the prerequisites are continuously changing [1]. Under these circumstances, in-depth understanding of software development requires being in midst of the situations as they are taking place. Doing research on related topics requires therefore close collaboration between the researchers and the practitioners. The need for close research collaboration is today understood ([2][3][4][5]) but practical ways to organize such collaborations are still challenging. As industry and academia leans on different governing variables [6] (e.g. publishing in academia vs. measure of market success in industry) it is important to find the practical ways to facilitate the needed close collaboration.

The software development industry has during the last decade aligned around agile methodologies [1], where practical methods around roles, ceremonies and artifacts facilitate stepwise growth of value with opportunity to adjust to changing requirements. The research performance in academia in software engineering has thorough research methods when it comes to studying phenomena and identifying problems, setting research questions, providing research results, and perform the validation, typically in an iterative way [7], but they have hardly a formally-defined process around timing, efforts and collaboration. In general, both industry and academia start with a problem or challenge that are not well understood, work in a continuously-changing environment, and the goals are achieved through iterative

steps. These similarities can be a fair motivation to ask the question *whether agile methodologies can be a good common ground for a successful collaboration in research between industry and academia*. Further, is it possible to apply certain agile practices established in industry, e.g. SCRUM? If yes, which adaptations should be introduced to which extent and for which type of collaboration? What is the dynamic of the deployment of the different agile practices? These are the questions that are scrutinized and discussed in this paper. The discussion is based on a longitudinal case study with several partners, both companies and universities.

This paper describes the deployment, the use, and the dynamic of utilization of SCRUM, the most widely used agile methodology in industry today. We report on a longitudinal industry – academia (I-A) research collaboration case, which has stepwise adapted SCRUM over a six-year period. We analyze the pace of its deployment, and its influence on the success of the projects. We also analyze a case where multiple research projects are performed in collaboration between several companies and research groups, and the effects of the synchronized performance in collaboration within and between the projects.

The paper is organized as follows. In section II we discuss the challenges of I-A collaboration, in section III we give an overview of our case study and define the research questions. In section IV we explain our research approach. Section V and VI describe collaborations in the case study within a project and between the projects, section VII discusses the analyzed data, and section VIII presents the findings and lessons learned. The study is concluded with the validity perspectives in section IX, related work in X, and conclusions in XI.

II. COLLABORATION CHALLENGES

The foundation for successful research between industry and academia collaborations lies in the possibility to create an overlap between the key activities and goals that both want to address. Industry is focused on identifying business critical problems and then address them with beneficial solutions. In parallel, academia is focused on identifying problems where there is a knowledge gap and then address these with new knowledge that contributes to the general body of knowledge. Figure 1 shows the research model (adopted from [7]) for industry and for academia. The two lines of activities, from industry and from academia, are indeed overlapping, but to make this overlap work in practice, a number of practices (i.e. collaboration ceremonies) need to be in place, which both parties agrees about and thrives over.

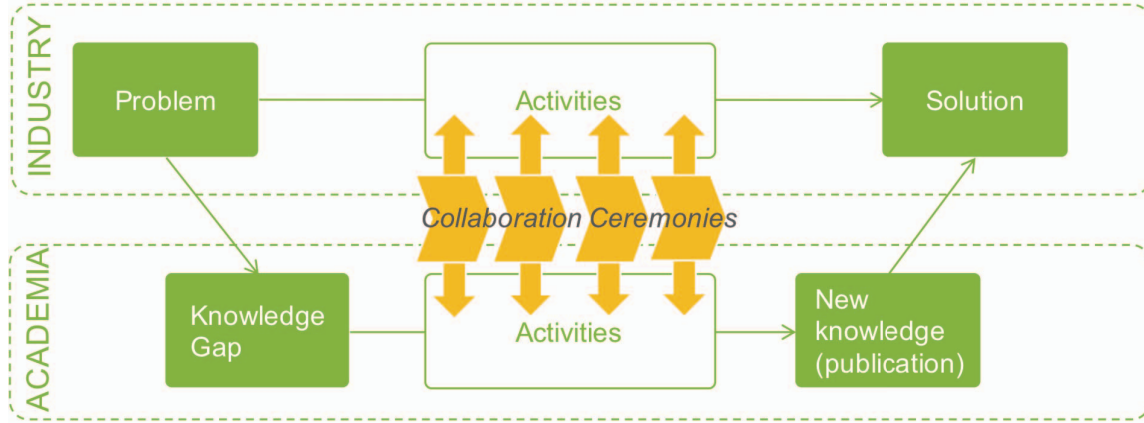


Figure 1. Overlap between industry and research activities

While it is relatively easy to agree on the problem and solutions, viewpoints differ regarding variables such as relevance, rigor, time horizons, and planning practices. Research has characterized these viewpoints also as success factors in collaboration management and software engineering research [5], but the practical methods to overcome the differences are still challenging.

One example of a collaboration challenge is *how to get started a research project*. Once having a good interest match between a researcher and practitioner [2], they have to agree about the research question(s). While the practitioner might not even see the need to define a specific question to meet the problem, the researcher is not satisfied with just a question matching the problem unless it also will contribute to new knowledge. Another example of collaboration challenge is about *how to work jointly during the period of understanding the problem*. The practitioner is used to a close collaboration where knowledge grows informally as the work evolves. The researcher is more used to formal procedures where data is collected and then analyzed possibly without any involvement from the practitioners during the analysis (for example when applying systematic literature study). A third collaboration challenge example is about *how to organize and execute the joint work* – different paces, deadlines and focused work is practiced in academia and industry. Finally, we can refer to a challenge *how to communicate with different stakeholders*. The practitioner is well aware about the need to share the development results continuously to the closest stakeholders to enable their fast utilization, but does not pay a high priority of sharing the results to boarder community. The researcher is often more used to share its result in workshops informally discussing preliminary results with other researchers that are not necessary close stakeholders, and later publish the results, which obviously require larger lead time.

While all these examples are indeed challenging when it comes to managing one collaboration initiative, there is another key challenge when to deal with many related research initiatives, i.e. scaling I-A research efforts such as

running several research projects between several industrial and academic partners. These initiatives need to have the attention from their stakeholders to assure the right involvement, commitment, approval and deployment. As the stakeholders now come both from both industry and academia, the joint access is difficult both from timing and time perspective.

III. THE CASE STUDY

The Software Center¹ is a consortium between academia and industry with its headquarter in Gothenburg Sweden, started in 2011 and during six years, the center has grown from seven to fifteen partners; ten from industry (Ericsson, Saab, Volvo Group, Volvo Cars, Axis Communication, Tetra Pak, Grundfos, Jeppesen, Verisure, and Siemens), and five universities (Chalmers University of Technology, University of Gothenburg, Malmö University, Linköping University, and Mälardalen University), gathering about twenty researchers and at least the three times more practitioners. The characteristics of all companies are that they are not pure software companies, but rather enterprises developing software-intensive systems, and facing increasing influence of software in development efforts and in providing novel features. The companies are classified as large international enterprises, with a high competence in system and software development, and accustomed to collaborate with academia partners in different research projects. The Software Center's mission is to in the light of the current rapid digitalization strengthen and transform software-intensive systems industry into modern world-leading software companies in terms of business models, architectures, ways of working and organizational setup. The Software Center has built its current successful progress on several different principles such as frequent partner interaction, transparency of outcome, close collaboration, industry funding, industry decisions, and academia research design. The research projects are grouped in four themes – continuous delivery,

¹ <http://www.software-center.se>

continued architecture evolution, ecosystems, and metrics. The research projects are executed in form on six-months long sprints, concluding with result reporting, and then the decisions from the steering committee whether the projects will continue, increase or decrease the efforts, or be closed down. Typically, several industrial partners are engaged in each project, and one research group, but it could be several. The projects are mostly proposed by the researchers, though it could be discussed or even initiated by industrial partners.

In our case study we pay the attention to fully understand the area ‘close collaboration’ realized through research projects that applied agile methodologies, concretely adopted SCRUM [8]. SCRUM is an iterative and incremental methodology widely used in industry, since it enables close face-to-face and online collaboration of team members, and it increases speed and flexibility in software development. SCRUM makes use of a number of roles and key ceremonies to reach its goals. Key roles are teams, scrum masters and product owners, while key ceremonies are ‘sprint planning’, ‘daily stand’, ‘sprint demo’ and ‘sprint retrospective’. The roles and ceremonies are used in a sprint workflow where a sprint is restricted to a fixed duration, normally between one to four weeks. Each sprint starts with a *sprint planning* event that aims to make an estimated commitment for the sprint goal. Each sprint ends with a *sprint demo* and *sprint retrospective* that reviews progress to show to stakeholders and identify lessons and improvements for the next sprints. The *daily stand* ceremony helps the team to respond to changes and navigate towards the goal. As SCRUM is designed to manage responsiveness, disciplines outside software development have started to use the it.

This model has been adopted in Software Center for collaborative research projects, with some changes: projects may last several years, a sprint is six months long, the number of teams are lower (3-4 teams), as well as the team sizes – 3-4 members). The results obtained each sprint are realized as values for industrial partners, and publications primary for the research partners. Figure 2 shows the SCRUM process and its key elements as a standard used in industry, and adopted in the Software Center.

Our study is focused on how the collaboration challenges described in section II are met in such constellation.

Concretely we analyze utilization and adoption of SCRUM, which we express with the following research question:

Q1. *What are the advantages and challenges in using SCRUM in collaborative research projects between industry and academia?*

In addition, we analyze the dynamics of deploying of SCRUM in an I-A collaboration, since while this process model is commonly used in industry, it is not known as a possible model for performing research activities. We define the following research question:

Q2. *What is a possible introduction pace of SCRUM in industry-academia collaboration?*

Finally, the case study enables us to analyze effects of using SCRUM in several collaborative research projects with many stakeholders. Here we state the question:

Q3. *Which synergy effects and which challenges are appearing in extended collaboration in the industry-academia research projects?*

These questions are discussed through a case study described below.

IV. THE CASE STUDY RESEARCH APPROACH

In the current study we applied Collaborative Practice Research (CPR) [9] that includes insiders (practitioners) and outsiders (researchers) working in close collaboration in a research team that can take advantage of bringing their specific knowledge of identification, analysis, and interpretation together. The two authors have had the roles as an insider and an outsider and then played the roles accordingly [10]. The insider has been an active partner to the studied case since it was started and has deep insights to the needs within in industry as well as the conducted projects. The outsider has joined the studied case three years ago with a long experience as a researcher, but with less detailed knowledge of the different projects. This set-up of roles made it possible to criticize and discuss relevant data. The authors took an iterative approach in the analysis phases to be able to scrutinize all data carefully. They both thrive over the topic of understanding how industry and academia can come closer

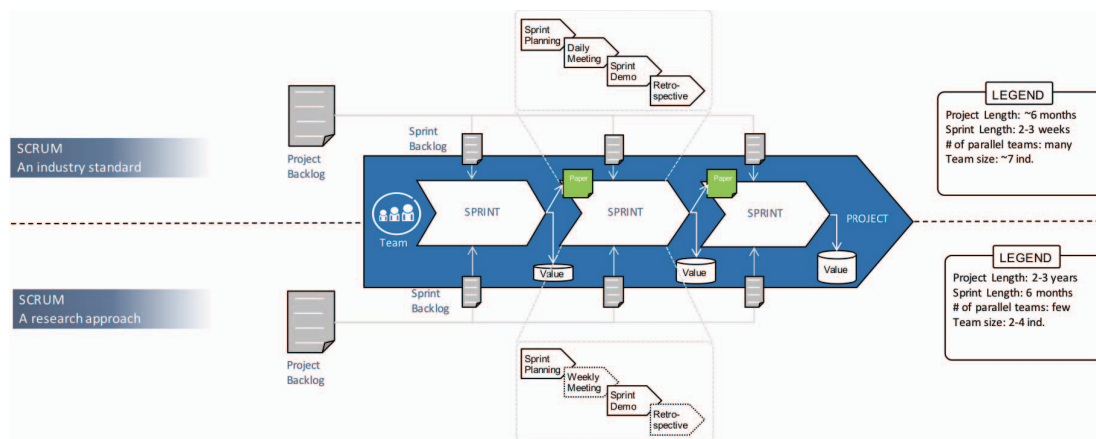


Figure 2. Key elements in SCRUM

and by that generate front-line research in context depending areas.

The CPR approach includes case studies ([11][12]) and this is the method used in this longitudinal study performed during two years, analyzing a period of six years. The information collection included several different methods and sources to triangulate data and avoid possible biased influence (see TABLE I). The insider has participated actively in the steering committee and project meetings, and had influence on strategic decisions and concrete activities, having mostly the perspective from the industry side. The insider was also included into projects works from time to time. The outsider performed the questionnaires and semi-structured interviews with the researchers and practitioners. During the study the insider and outsider had regular meetings discussing the results in several iterations. In total three questionnaires have been performed, two with the researchers (2015, and 2016), and one with the industrial participants (2016). The questions to the researchers were related to researchers' satisfaction of the collaboration and research results, while questions to the practitioners were related to the research result relevance.²

TABLE I INFORMATION ELICITATION

Form and source	Action
Direct involvement	Directly involvement in ~15 research projects and in the Steering Committee.
Minutes of meetings	Formal notes on discussions and decisions made during governance meetings.
Participatory observations	Taking part of action where problems are discussed and analyzed in different projects.
Development documentation	Analysis of project meeting protocols, project presentations, project plans, project decisions, final reports.
Collaboration questionnaires	Two questionnaires about research appreciation of the result and one questionnaire industry about appreciation of the way of working.
Open-ended, semi-structured interviews	10 informal interviews or discussions with practitioners and researchers based on the questionnaires.
Workshops and analysis meetings	2 workshops with discussions with researchers and practitioners about the results from the questionnaires.

V. COLLABORATION ANALYSIS

Table II shows the I-A collaborative projects during the history of Software Center. The table shows the number of concurrent projects, number of partners involved in them, the size of the project in terms of active members, and activities in form of workshops. Further, the number of the characteristics of SCRUM, i.e. ceremonies are presented: sprint planning, monthly stands, demonstrations, and retrospective activities. Finally, the table shows the outcome in form of research results in from of publications, and the results taken over from industry.

When analyzing the Software Center data from Table II, and from of other data sources as specified in Table I, between 2011-2016 we can identify that the collaboration has evolved in two major steps. The first phase, *setting the pace*, occurred from the start of the Software Center in 2011 until mid-2013. The second phase, *extending the use of collaboration ceremonies*, occurred from mid-2013 until 2016. Below we describe the two phases in more detail.

A. Phase 1- Setting the Pace

Already from the start it was obvious that the Software Center should be built on principles of close collaboration. The center early defined one important principle: 'working in 6-month sprints'. The idea was that all research collaboration projects are defined and reported within a 6-month time frame to assure stepwise and visible progress that provided continuous interest from stakeholders. Initially, researchers and practitioners came together in specially organized events where the practitioners described their software development challenges around a number of themes (e.g. architecture and metrics). Researchers then approached the practitioners where they wanted to continue discuss potential ideas. In this way the different collaborations were initiated.

Each researcher (or researcher team) then proposed a research project for the upcoming 6-month sprint based on the joint ideas from the researcher and collaborating industry partners. All incoming proposals were refined and decided upon in an organized governance structure where the industry partners decided upon what proposals they wanted to finance. In the end of each sprint all collaborating partners and interested stakeholders came together in a major reporting workshop, where all research project results were presented. The same workshops were used to discuss both evolving and new research projects for the next upcoming sprint.

Setting the pace with 6-month sprints has shown to be beneficial. The joint sprint planning by every research project as well as the joint demo in the end of every sprint, facilitate the close collaboration. The sprint planning ceremony assures cross industry and academia agreement as well as a plan for how to reach a specific goal. During the demos (e.g. reporting workshops), which today gather around 100 different collaboration partners from both industry and academia, the results are widely presented and disseminated. These two very important collaboration ceremonies form the basis for a continuous close collaboration where skilled people within different disciplines can come together and solve intricate challenges for the future.

This way of working faced also some challenges. Researchers report that the sprint concept can be challenging for the researchers due to time limitations to utilize resources on short notice as well as having time to report on the study. Also, industry partners have identified challenges related to the time limitations where there are relatively short periods (one week) between project proposals are available until they should be decided about.

² the questionnaires and answers can be found at: www.goo.gl/P2c867

TABLE II. RESEARCH PROJECTS WITHIN SOFTWARE CENTER 2011-2016

Basic Data						SCRUM Ceremony Usage (All = 100%, Many => 50%, Few < 50%)				Outcome	
#	Sprint	# of research projects	# of SW Center partners (I-A)	Average # of research project members	Average # of collaboration workshops per research project	Sprint Planning	Monthly Stand (theme coordination meeting)	Demo	Retrospective	# of publications - # of industry partners using at least 1 of the project's outcome	Example of practical project outcomes (*)
1	Fall 2011	7	4 - 2	4	3	All	None	All	Few	~10 - all	Measuring Size of Changes
2	Spring 2012	8	4 - 2	5	3	All	None	All	Few	~15 - all	The Stairway to Heaven Model
3	Fall 2012	8	4 - 2	5	4	All	None	All	Few	~20 - all	The CIVIT Model
4	Spring 2013	10	4 - 2	6	3	All	None	All	Few	~25 - all	The CitiM Model
5	Fall 2013	11	7 - 3	6	3	All	All	All	Few	~25 - all	The Hypex Model
6	Spring 2014	12	7 - 3	7	4	All	All	All	Many	~30 - all	The Caffea Framework
7	Fall 2014	14	7 - 4	7	4	All	All	All	Many	~30 - all	Self Healing Systems
8	Spring 2015	15	7 - 4	7	3	All	All	All	Many	~35 - all	The TeLESm
9	Fall 2015	17	7 - 5	6	3	All	All	All	Many	~35 - all	The Stairway to Heaven 2.0
10	Spring 2016	20	10 - 5	6	3	All	All	All	Many	~35 - all	The Data Sharing Model

(*) THE PROJECT OUTCOMES: [HTTP://WWW.SOFTWARE-CENTER.SE/DIGITALASSETS/1580/1580347_SW_CENTER_DELIVERABLES_REVB.PDF](http://www.software-center.se/digitalAssets/1580/1580347_SW_CENTER_DELIVERABLES_REVB.PDF)

B. Phase 2- Extending the use of collaboration ceremonies

By time the number of research projects grew (see Table II column 3). To handle scale, it was needed to organize the different research projects that were closely connected. As the Software Center was formed around several themes, these were then used to get collaborating partners in different projects together. This way of organizing the projects also contributed to having stronger research projects (i.e. more precisely defined research questions, industry partners and researchers) to be able to come up with stronger results (more data sources and more thorough analysis). These theme coordination meetings (i.e. Monthly Stands) filled the function of regularly coming together to respond to changes and handled issues during execution towards the set-up goals. By time, the potential of learning and doing better in the next sprint became attempting to all involved. One of the founding professors already from the start practiced mini-retrospectives in the end of every sprint demo, to be inspired from potential improvement ideas. This helped evolve the Software Center by understanding how to organize demo events in ways even more attractive for all stakeholders. Gradually more and more research project practiced retrospectives in the end of every sprint to do better in the upcoming sprint (see Table II, column Retrospective).

Making use of monthly stands and retrospectives have increased both the collaboration opportunities and the improvement opportunities. Finding ways to regularly come together to solve a challenge or get better at something is very important for stakeholders (i.e. practitioners and researchers) who does not meet normally in the every-day business life.

It is still challenging to get many researchers to gather around the same research question and the organize the research accordingly. The current collaboration ceremonies help researchers come together in individual projects, but so far there are few signs of projects where more than a few researchers join forces to understand larger development challenges cross many companies. It is likely so that other practices are required to enable further cross-project collaboration where researchers and practitioners can come together to discuss potential broader projects.

C. Overall acceptance of SCRUM

In two questionnaires with the researchers the overall goal was to understand appreciation (or not) of using SCRUM and in particular sprints of six-months periods. For the question about appropriateness of the Sprint model the answers were in majority positive with 89% of answers “good, very good, excellent” (see Figure 3.).

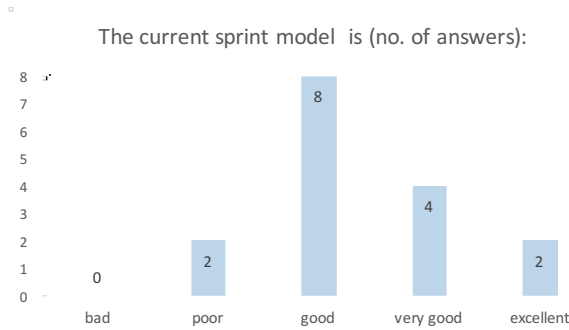


Figure 3. Appropriateness of Sprint model

The most characteristic positive answers were:

- "The short period grants companies' interests, which are more open to take a 6 months risk."
- "The sprint model is critical for frequent feedback and validation of results. Also, it allows a healthy redirection of projects if necessary and it stimulates research results in a very positive way. Finally, it is great fun to have report results every sixth months as well as to lick off something new every sixth months. The rhythm it brings to the research process in unique and very rewarding."
- "Good for keeping industrial relevance."

The cautious and negative answers (despite dominated positive grading) were:

- "... however, the sprint without a corresponding backlog (i.e. a concrete long-termed plan agreed by the companies) is not so useful."
- "Reporting results frequently have benefits but development, execution and reporting of results every 6 months puts, in my own view, limitations on what type of research that can be done."
- "... bad for long term research goals with e.g. involving PhD students."
- "It is good, but expectations are high and unclear."

The second important and to some extent a controversial element in the SCRUM process has been shown to be the decision process for the project start and its possible termination. The decision is given by a body of industrial partners based on a short proposal written by the researchers and already informally discussed with the industrial partners. These decisions are of course very important, not at least for the researchers, which mean for them new projects funded or not. The question about the project decision process was stated as a question whether the decision process is good or not for the research of the academia partners. Figure 4 shows the answers. The majority is neutral or positive, while two researchers (of 15) had a negative opinion.

The answers explained in a positive manner:

- "It forces us to formulate projects with clear and concrete benefits for the companies, which is good."
- "Companies get to see what is available and commit resources to the projects they like. As such the process also serves to weigh the value of the presented research."
- "Projects should be frequently evaluated to stay relevant. Also, the selection process allows very useful feedback

from the companies as they all engage in the process, read the project descriptions and share their thoughts with us in an impressive way."

The negative comments were the following:

- "At now I honestly did not understand how the selection works, nor could grasp from the companies if our proposal was appealing or not."
- "About the selection process itself, I believe there is not always clear relation/correspondence or there are even discrepancies between: how well a project is doing, in terms of topic and results relevance and impact for the company partners, and the feedback/decision from the selection process."
- "About the selection process itself, I believe there is not always clear relation/correspondence or there are even discrepancies between: how well a project is doing, in terms of topic and results relevance and impact for the company partners, and the feedback/decision from the selection process."
- "Not transparent, frustrating for researchers and industry partners. Facilitates rumors."

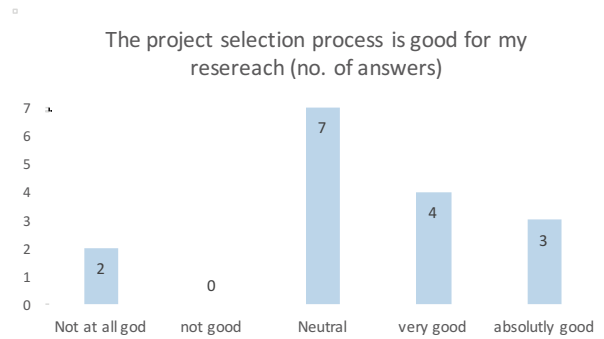


Figure 4. Satisfaction with the project start decision process

Here we see that the process itself is not questioning but rather the transparency of the decisions, and the rationale given for the decisions. (After the presentation of the answers, the Steering Committee decided to increase the transparency of the decision process).

The other questions we refer to here are related to the industrial relevance and the importance of the work for the researchers. The answers show if the SCRUM approach can be used for the research that is important for all partners. The summary of the answers is presented in Table III:

TABLE III THE RESEARCH RELEVANCE

Question	Answer
Is the research in Software Center important for you?	Yes 70% (4 and 5 in the scale 1-5), 30% neutral
Is Software Center important for increasing your network?	Yes 75% (4 and 5 in the scale 1-5)
Is the research at Software Center for you of the high excellence?	Highest 18%, High 35%, OK 35%, Low 6%, Poor 6%

We note here that we did not analyze the scalability of the projects and interrelations between them. The projects were

basically running independently, and during the workshops some mutual activities have been initiated, but not in a systematic way.

VI. EXTENDED COLLABORATION

The six-month-sprint-approach enables synchronization of several projects running in parallel and executed by different companies, research groups, and universities. When using a six-month-sprint-approach combined with a number of collaboration ceremonies, more collaboration opportunities occur apart from the ones in the pure research projects. Several positive collaboration synergies reveal once the pace of ceremonies are regularly executed. While the initial Software Center ambition was to increase the excellence in I-A research projects, several different levels of collaborations occur that are making use of different collaboration ceremonies. While the most obvious collaboration level is the one in the research projects (see Figure 5, #2) where practitioners and researchers come together, five other collaboration levels also seem to be of high value. Leaders from industry and academia come together (see Figure 5, #1) in classical governance meetings and sprint demos, where they learn to appreciate each other's strengths and respect their different governing variables. Practitioners from different companies come together (see Figure 5, #3) to listen and learn from each other from the 'horse's mouth', where they get practical guidance and understanding of the same challenges from new perspectives. Researchers from different universities (see Figure 5, #4) come together to discuss and evolve ideas, where joint interests can pave the way for future stronger research projects. Leaders and practitioner within one company (see Figure 5, #5), particularly large and/or distributed companies, come together and learn that company internal needs often are the same and can be improved by jointly coming together. Researchers within one university (see Figure 5, #6), particularly large universities, come together via the different collaboration ceremonies, where they learn more about each

other's research topics and increase spontaneous new ideas. The different collaboration levels make use of different collaboration ceremonies (Figure 5, columns 3 – 8). This increase of collaboration between different stakeholders within industry and academia is in itself beneficial for the in-depth understanding of all aspects that are required to accelerate the adoption of novel approaches to software development. Again, the practical ways (i.e. the collaboration ceremonies) to meet and collaborate is key to assure continued, increasing and successful research projects.

When studying the company presence at the quarterly steering committee meeting we found that almost 90% of the industry partners took part at every meeting. During six years one company left the center, due to closing the business, all others stayed and new joined in spite of variation in market position. However, as the center is growing there are challenges related to scalability. From the interviews, we have learned that researchers claim that it is very difficult to manage to present and discuss deeper insights during the 10-minutes slot given at the demo ceremony. At the same time, just extending the time for the demo requires too much total time that many participants are not ready to give.

VII. DISCUSSION

A. The positive effects of SCRUM

The need for time-boxed and coordinated sprints to manage speed and responsiveness are similar. The sprint length differs, but the beat in itself where all collaboration ceremonies can be executed in a sufficient pace is as valuable in industry as for I-A research projects. The foundation of having teams consisting of both researchers and practitioners that work as a unit towards a joint goal is similar. In the case of Software Center, the team members, in particular the researchers, most often are allocated for more than one project. This fact underlines the importance for synchronization between the projects. The sprint planning ceremonies meet the challenge of agreeing about what to do, how and when in order to reach the goal. The challenge of regularly

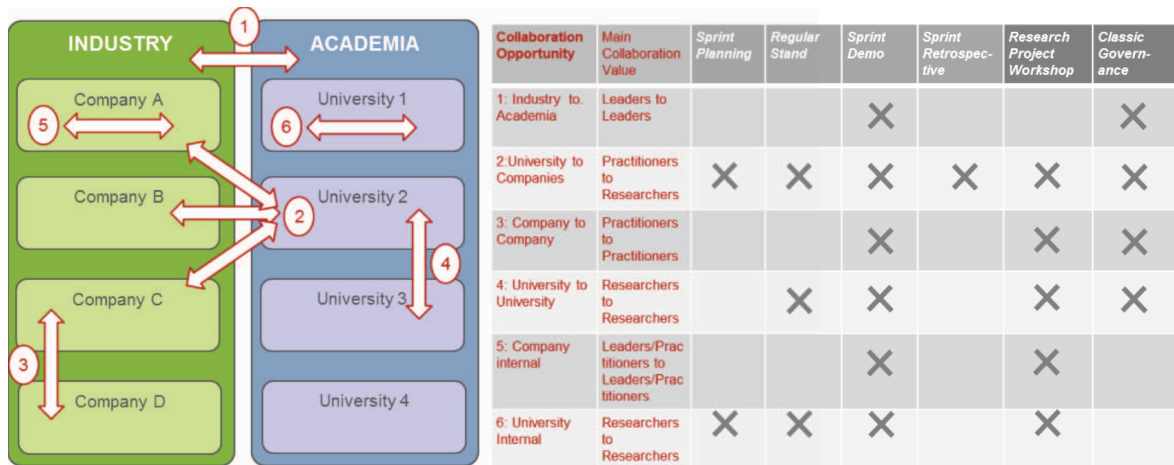


Figure 5. Different Collaboration Levels

coming together to mitigate practical problems during executions are the same, even though they are less frequent (monthly stand) in I-A research project compared to industry projects (daily stand). The challenge of having their stakeholders updated and interested is the same and is met with demos, but again with different frequencies. And finally, anything that can be repeated can be improved with retrospectives. This statement is as valid for I-A research projects as for pure industry development projects where retrospectives are most useful to learn and do better.

B. The challenging parts of SCRUM use

There are however several challenges that were repeatedly appeared during the projects executions. When discussing with the researchers a number of issues were brought up. The most common ones are related to the 6-months sprint length (too short), researchers' availability (too little time) and researchers' motivations in doing demos (since it can require quite some efforts, that is not visible as a real research contribution). The researchers that were a part since the start of the Software Center, and also in similar research collaboration prior to that, do not see these issues as particularly challenging, so it is obvious that overcoming these challenges are related to how used you are to work in this fashion. Furthermore, almost none of the challenges are directly related to the collaboration ceremonies. Other challenges that are brought up by the practitioners are related to difficulties of making the research projects stronger and having publications closely connected to research results.

C. What was not used from SCRUM

While most of the elements from SCRUM have been used and possibly adopted in the Software Center I-A collaboration, there are several elements that have not been used, either on purpose, or so far.

Product owner (product owner creates a prioritized wish list called a product backlog) has not been explicitly defined. In the projects it was assumed that the main (senior) researcher is the product owner. However, this role is not used on purpose due to a strong interest for the results of both industrial and academic partners, and not precisely defined product backlog. Having in mind that these are the research projects, it has been acceptable that each sprint does not need to start with the precisely defined product backlog.

Product backlog (a list of things that need to be done in the project) is for the same reason not required. Some of the projects used product backlog, but in a more informal and non-transparent way. An absence of a product backlog may however cause some uncertainties in the projects (as we have seen in a few comments in the questionnaires.)

SCRUM master (accountable for removing impediments to the ability of the team to deliver the product goals and deliverables) is not defined in the Software Center projects. It is assumed that this is the main researchers (so the same person as the product owner) which might have some negative consequence on the project work efficiency, but so far a need for an explicit role in the projects has not been identified.

D. The evidence of the research results

The results provided by the center in a form of values for the companies are shown by a continuous increase of projects and partners. The results in the form of the research achievements, in particular in number of publications, are not presented in this paper, but the internal sources list publications of more than 200 research papers (including peer reviewed workshop, conference papers and journal articles), which shows at least a high publication productivity. The list of artifacts transferred to the companies indicate also a high acceptance (see TABLE II), and a high value perception of the research results. This perception is confirmed by the industrial partners by their answers in the questionnaire. This perception also addresses the general criticism around I-A collaborations about lack of research relevance [13]. The usage of the results within industry has often been vague.

VIII. THE FINDINGS

Finding practical ways to come together for collaboration is necessary to facilitate successful I-A research. The need to be responsive towards the business environment and act with speed is as vital in research as in development. When pace and requests on the development changes, so does the pace and requests on researching development. Research set-ups striving for close collaboration between industry and academia can benefit from taking the following findings which we list in a form of recommendations.

1. *Organize research in sprints.* Research projects organized in six months sprints tend to be attractive for both industry and academia.
2. *Utilize collaboration ceremonies.* SCRUM ceremonies help researchers and practitioners to frequently and regularly meet and communicate.
3. *Appreciate getting close.* See all collaboration ceremonies as opportunities for input to even more research rather than cumbersome time consuming events that prevent the research.
4. *Plan the research jointly.* Start the research work by coming together and plan the sprint. Assure a jointly needed research question that opens up for mutual activities and understandings.
5. *Find the pace for reconciliation.* All collaborations need reconciliations. In development, a daily 15 minutes stand is recommended. In research contexts, meeting up weekly or monthly is maybe more in line with a 6 months sprint approach.
6. *Thrive over accomplishments together.* Having demos where everyone is invited to listen to everybody creates an urge and willingness to create more success jointly.
7. *Learn continuously from previous sprints.* Any activity that is repeatable is an object for being improved. Six months sprints is an excellent activity to do retrospective on to do even better in next sprint.
8. *Look beyond the pure research project.* Every opportunity where leaders, researchers and practitioners come together is an opportunity to maintain and grow ongoing and upcoming research projects.

9. *Scale up!* Increasing number of projects and number of stakeholders (companies and universities) running in parallel sprints increase the outcome.

A typically successful research project within this case study utilize all the benefits of the identified collaboration ceremonies, while they at the same time have found a distinct match between the problem and the knowledge gap. They tend to plan thoroughly from the start, so that participation all project member can participate and contribute as plan.

IX. VALIDITY

In a longitudinal case study [14] there are several threats to validity; we analyze constructive validity, internal and external validity, and the reliability of the research results.

The constructive validity can be jeopardized by lack of relevant information, in particular from the past. This threat has been managed via three different measures: insider/outsider roles, triangulation and iterative analysis. The access to the internal documents of different types, such as minutes of meetings from the projects and steering committee meetings, the documented project results, in a combination with surveys and interviews compensated the problem of variation of information accuracy during the period of six years. The outsider role helped in analysis of the collected data. The analysis was done in two iterations followed by three surveys, and the interviews as feedbacks to the survey results.

The internal validity is threatened by the changes that occurred during a longitudinal study where individuals moved, changed tasks, made careers, resigned etc. For this reasons the recent data collected during the case study period (last two years) are more accurate, due to more available sources. Since the character of work did not changed radically but rather evolutionary during six yes, it was possible to assume that similar conditions were valid during the entire period and that no hidden variables, that later disappeared, determined the results. While one can expect that the initial period was characterized more by the variation in project and collaboration work, it was still possible to extract relevant data for the analysis such as efforts from the researchers, number of the projects, the results provided, which gave enough accurate input for the analysis.

The external validity, i.e. generalization of the result, is always challenging in context specific research approach and here the time-period make it extra challenging. The generality in respect to the domain is ensured since companies from different business and engineering domains, as well as researchers from different universities were involved in the projects. The geographic distribution might threaten the overall generality, although most of the companies belong to international enterprises, the project participants were from the local business units, and certainly had the Scandinavian culture in which a cooperation between industry and academia is quite established phenomenon, and in general is functioning very well. That certainly helped in collaboration. In a similar way, a tradition of flat organizations, direct

contacts, and shared responsibilities contribute to the successful and effective collaboration. The behavior of companies and universities could be different in other geographical areas. In a similar way, we cannot generalize the results for SMEs, though SMEs are usually more flexible in collaboration than large international companies, so we could expect similar results. Further, we can state a question whether any type of research would show similar behavior. The research at Software Center is relayed to architectural concerns, integration and deliver issues, and software evolution. Possibly another type of research, e.g. constructive, may require some adaption in form of long-term goals. Finally, we can ask whether the collaboration ceremonies are suitable only for I-A research or could they also be applicable for any kind of research project? Our case study only provide data for the first, and for specific types of the projects.

The reliability of this study was easier to ensure, since the data collected are directly related to the results, and data were available from different sources, including internal and the public available data.

X. RELATED WORK

In [15] some of the results from Software Center are presented, and there sporadically the working methods are discussed, but not in a systematic way, and not focusing on the research methods. There is a plenty of literature about agile methods and SCRUM, their application and adoptions in different contexts in industry [16][17][18][19]. There is, however, very little evidence in the literature about utilizing agile methodologies or principles in I-A collaboration. In a previous work of one of the authors [2] a set of best practices in a I-A collaboration are identified, including those related to support the collaboration (such as addressing activities that ensure the result, communicating the process and the results, and funding small projects), and those that are based on the agile principles to be applied in research (such as organizing “get-togethers”, and attending to concrete needs and goals). In [20] several I-A collaboration projects using agile methods are analyzed showing which practices are mostly used (e.g. acceptance testing, small releases, sustainable pace). There is a large body of literature describing successful (I-A) collaboration in the domain of software development, and the findings are sometimes related to use of agile principles. In [21] success factor in an I-A collaboration are discussed, and those related to agile principles are “short term results and impact on industry”, “researcher has a visible presence in industry”, and “regular meetings”, and in [22] these findings are empirically evaluated. In [23] it was shown that even the successful I-A research collaboration, with findings and the results that are useful for industry are not enough for a successful technology transfer. In his blog³ [Finkelstein](http://blog.prof.so/2013/01/collaboration.html) identifies “golden rules” for a successful collaboration, one of them is “share strategy and timelines”. The absence of evidences of using agile methods in I-A collaboration calls for empirical analysis of successful collaboration and identification of practices related to agile principles.

³ Anthony Finkelstein, The Golden Rules of Successful University Industry Collaboration, <http://blog.prof.so/2013/01/collaboration.html>

XI. CONCLUSION

In this paper we have discussed three questions related to use of SCRUM methodology in I-A collaboration research projects. The three questions (Q1: What are the advantages and challenges in using SCRUM in collaborative research projects between industry and academia?, Q2: What is a possible introduction pace of SCRUM in industry-academia collaboration?, and Q3: Which synergy effects and which challenges are appearing in extended collaboration in the industry-academia research projects?) were answered through a longitudinal case study. The case study has shown a successful adoption of SCRUM in I-A collaboration. There are several factors contributing to this success that was shown: Successful research between industry and academia lies in the possibility to create an overlap between the key activities that both partners want to address. Embracing collaboration ceremonies facilitate I-A research collaboration. Frequently and regularly coming together is key for being responsive towards upcoming requests and this responsiveness increases both quality and speed of the deliverables. The trust between the collaborating partners increases and by that, a willingness to continue the collaboration over time. We have also seen that the activities synchronization between the projects provide positive effects on the knowledge sharing and technology transfer, increase the acceptance of the achieved results, and in general increase the enthusiasm to continue and extend the cooperation.

The presented case can further be studied to provide new evidence related to the type of the research projects and the relevance and excellence of the research results. Additionally, roles of the researches and the industrial partners in relation to the product owner and scrum master can be analyzed. Also the scalability factor can be studied – how much partners and research group scale up in the collaboration? What would be the optimal number in respect to achieved results in short and long term. It is also likely that other principles beyond SCRUM can be beneficial for I-A collaborations, but these are still to be researched. Finally, a question whether SCRUM (or other agile methodologies) is suitable as a research approach for other types of research, so not only for I-A collaboration, might be further studied, as it may lead to new forms of research.

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